

Wavelength Selection for Pepsin Collagen ME-HSI Data

Supplementary Experimental Report

Abstract

We apply our deep learning wavelength selection framework to a pepsin-digested collagen dataset acquired with Multi-Excitation Hyperspectral Imaging (ME-HSI). The dataset comprises three crosslinker concentration levels (0.002%, 0.0075%, 0.013%) imaged across 6 excitation wavelengths (310–400 nm) with emission recorded from 420 to 720 nm (10 nm step), yielding 158 excitation-emission band combinations per pixel. Using the autoencoder-based perturbation analysis method developed for lichen samples, we evaluate wavelength selection on this chemically subtle dataset where class differences arise from crosslinker concentration rather than material composition.

Results demonstrate that band selection consistently exceeds the full-spectrum baseline: KNN classification with 30 selected bands achieves 85.6% accuracy versus 79.8% baseline (+5.8 percentage points, 81% band reduction). When evaluated with Linear Discriminant Analysis (LDA), just 5 selected bands achieve 84.6% accuracy — exceeding the KNN baseline by 8.3 percentage points with 97% data reduction. All 10 tested classifiers match or exceed their own baselines at 30–50 bands, providing classifier-independent validation of the selected wavelengths. These results confirm the framework's generalization to new sample types and demonstrate that the selected bands capture the essential spectral information for discriminating subtle chemical differences in biological tissues.

Dataset Description

The Pepsin Collagen dataset acquired with the ME-HSI system with 6 excitation wavelengths. Collagen sponges with three crosslinker concentrations were arranged in a 3×3 grid (3 concentrations × 3 replicates). The dataset was preprocessed with exposure time and lamp power normalization using instrument metadata, which corrects for the 100× variation in exposure times across excitation wavelengths (310 nm: 1407 ms vs 385 nm: 14 ms).

Parameter	Value
Excitations	310, 325, 340, 365, 385, 400 nm
Emission range	420–720 nm (step 10 nm)
Total bands	158 (22–31 per excitation)
Spatial dimensions	256 × 348 pixels, 39,970 valid

Methodology

The wavelength selection pipeline follows the three-stage framework described in the main paper: (1) a 3D convolutional autoencoder with parallel excitation branches learns a unified latent representation, (2) perturbation-based attribution traces reconstruction sensitivity to individual excitation–emission pairs, and (3) Maximum Marginal Relevance (MMR) selection balances informativeness with spectral diversity. The autoencoder was trained for 30 epochs on the normalized pepsin data. 432 configurations were evaluated, spanning 9 band counts (5–130), 2 dimension selection methods (PCA, variance), 2 perturbation methods, 3 normalization methods, and 2 magnitude variants.

Additionally, 10 classifiers (KNN, SVM, Random Forest, Gradient Boosting, LDA, MLP) were evaluated to assess whether the selected wavelengths generalize beyond KNN. Each classifier was tested with both its full-spectrum baseline and the autoencoder-selected bands at 8 band counts (3–50).

Results

KNN Pipeline Performance

Figure 1 shows the accuracy envelope across all 432 KNN configurations. Unlike the standard expectation that accuracy monotonically increases with band count, the Pepsin dataset shows a peak at 30 bands (85.6%) followed by a decline. This inverted-U shape indicates that the full 158-band spectrum contains noise that actively harms KNN classification, and the band selection method successfully identifies and removes these detrimental bands.

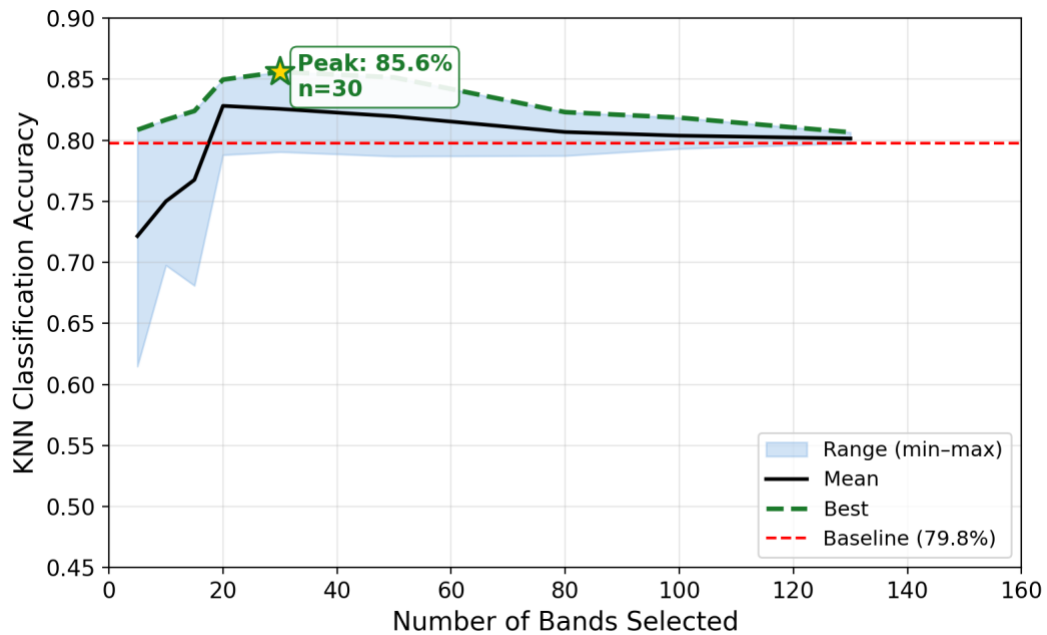


Figure 1. Accuracy envelope across 432 KNN pipeline configurations. All tested band counts exceed the baseline (red dashed line). Peak accuracy of 85.6% occurs at 30 bands (81% reduction).

Figure 2 compares the best KNN result at each key band count against the baseline. Green bars indicate configurations exceeding the baseline.

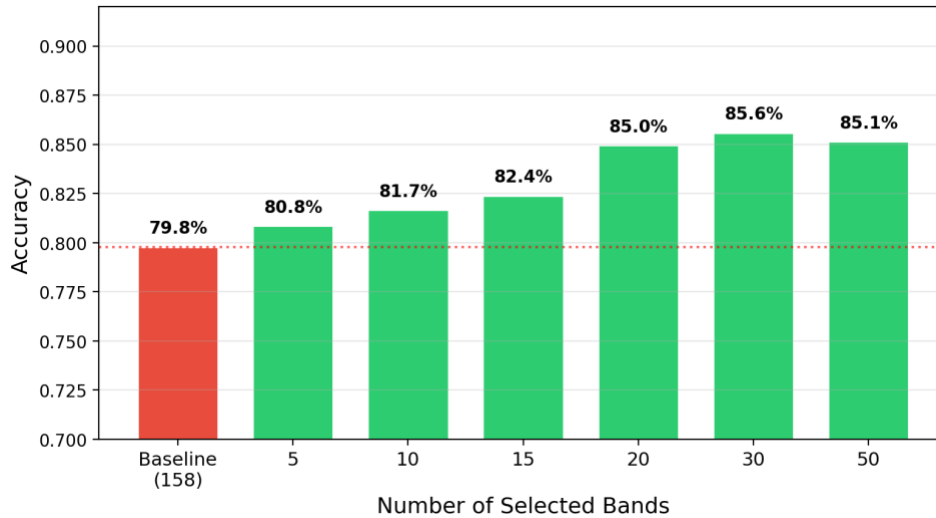


Figure 2. KNN classification accuracy at key band counts. Green bars exceed the all-bands baseline (79.8%). Every band count from 5 to 130 surpasses baseline performance.

Summary of Key Results

Bands	Reduction	Accuracy	F1	Kappa	Configuration
158	0%	79.78%	78.89%	0.6961	Baseline (all bands)
5	97%	80.85%	80.83%	0.7125	bands_5_pca_dim_1_perc_mag_medium_non
10	94%	81.67%	81.14%	0.7248	bands_10_pca_dim_3_abso_mag_medium_max
15	91%	82.39%	82.19%	0.7362	bands_15_pca_dim_3_perc_mag_medium_non
20	87%	84.96%	84.81%	0.7745	bands_20_var_dim_1_perc_mag_medium_max
30	81%	85.59%	85.44%	0.7838	bands_30_var_dim_1_perc_mag_medium_max
50	68%	85.14%	84.90%	0.7771	bands_50_pca_dim_3_perc_mag_medium_non
80	49%	82.30%	81.83%	0.7343	bands_80_pca_dim_1_perc_mag_medium_max
100	37%	81.85%	81.40%	0.7274	bands_100_pca_dim_1_perc_mag_medium_max
130	18%	80.65%	79.99%	0.7092	bands_130_pca_dim_1_perc_mag_medium_max

Table 1. Best KNN (k=5) accuracy per band count. All entries exceed the baseline.

Multi-Classifer Validation

To verify that the selected wavelengths are informative beyond KNN, we evaluated 10 classifiers on the same selected bands. Figure 3 shows accuracy curves for key classifiers. LDA achieves 84.6% with just 5 bands and 92.5% with 50 bands, dramatically exceeding the KNN baseline of 76.3%. The LDA baseline with all 158 bands is 92.8%, so 50 selected bands retain 99.7% of LDA's full capability.

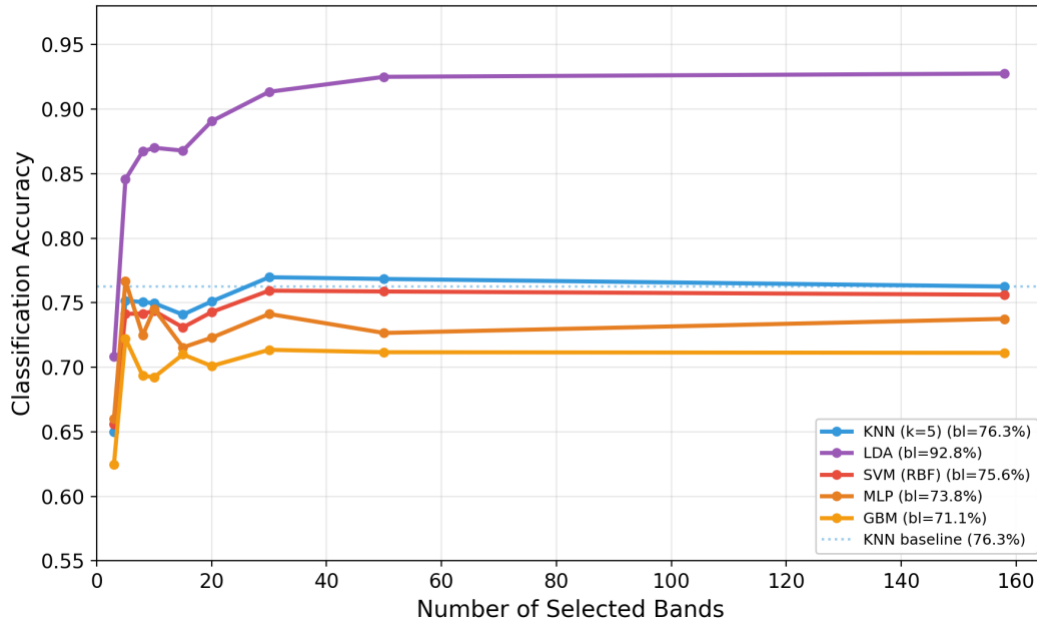


Figure 3. Classification accuracy vs. number of selected bands for six classifiers. LDA dominates at all band counts. Dotted blue line shows the KNN baseline.

Classifier	Accuracy	F1	Kappa
LDA	92.75%	92.74%	0.8911
KNN (k=11)	76.28%	74.89%	0.6432
KNN (k=11, dist)	76.28%	74.89%	0.6432
KNN (k=5)	76.26%	74.81%	0.6429
SVM (Linear)	75.82%	74.07%	0.6362
SVM (RBF)	75.62%	73.80%	0.6333
MLP	73.75%	72.04%	0.6057
RF (300)	71.16%	68.77%	0.5661
GBM	71.12%	68.39%	0.5659
RF (100)	70.19%	67.91%	0.5513

Table 2. Classifier baselines using all 158 bands.

Fair Comparison: Each Classifier vs Own Baseline

Figure 4 shows the gap between each classifier's selected-band performance and its own full-spectrum baseline. Green cells indicate the classifier matches or exceeds its baseline; red cells indicate degradation. Notably, all classifiers achieve green at 30–50 bands, confirming that the wavelength selection is classifier-agnostic.

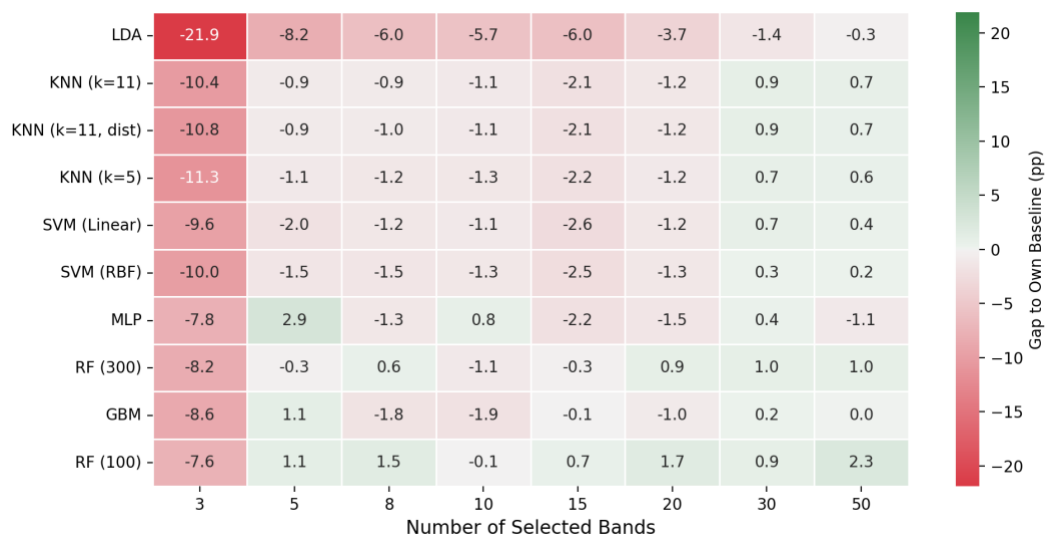


Figure 4. Gap to own baseline (percentage points) for each classifier at each band count. Green = exceeds baseline; red = below baseline. All classifiers match at 30–50 bands.

Selected Wavelength Distribution

Figure 5 shows the mean importance score across the excitation–emission grid for configurations that exceeded the baseline. The 365 nm and 385 nm excitations dominate, which aligns with the collagen fluorescence characteristics. After exposure and power normalization, these excitations reveal the strongest fluorescence signals and the most discriminative spectral features between crosslinker concentrations.

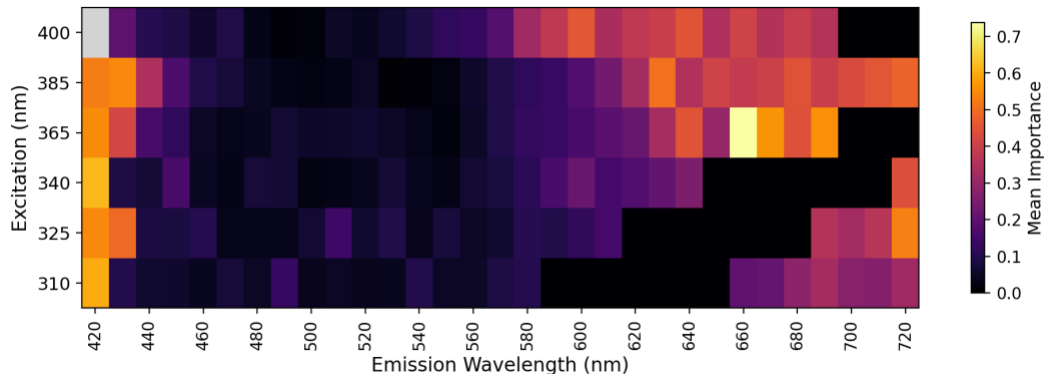


Figure 5. Wavelength importance heatmap across the excitation–emission grid. Gray cells indicate physically invalid combinations (emission < excitation + cutoff).

Band Reduction Efficiency

Figure 6 shows the accuracy retained (relative to each classifier's own baseline) as a function of band reduction percentage. LDA retains 91% of its performance even at 97% band reduction (5 bands) and 99.7% at 68% reduction (50 bands). KNN and SVM show steeper degradation, reaching their baselines only at 80% reduction.

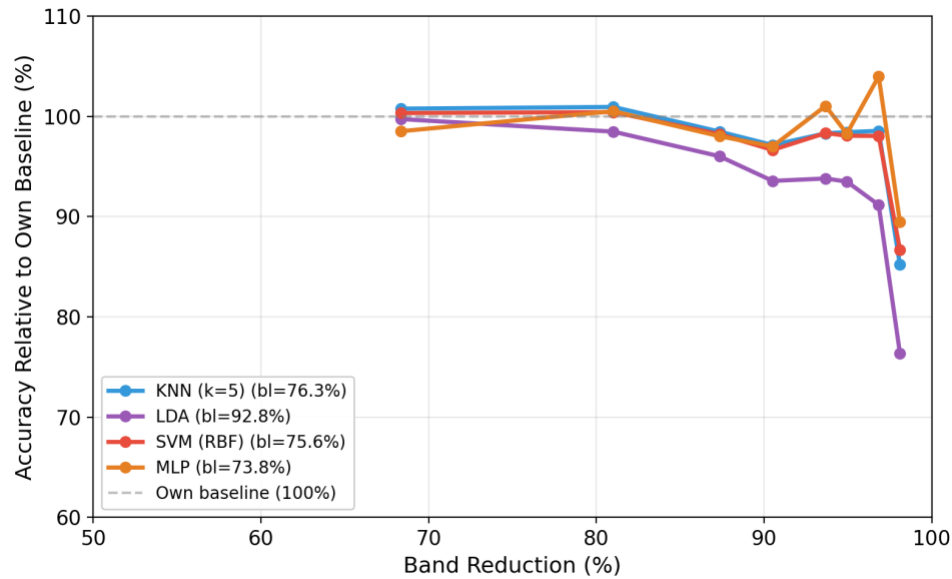


Figure 6. Accuracy relative to own baseline vs. band reduction for key classifiers.

Parameter Sensitivity

Figure 7 shows the effect of the three main pipeline parameters on accuracy. Both PCA and variance-based dimension selection perform comparably on this dataset, with variance slightly better at 20–30 bands. The "none" normalization method consistently outperforms alternatives. Percentile and absolute range perturbation methods show similar performance profiles.

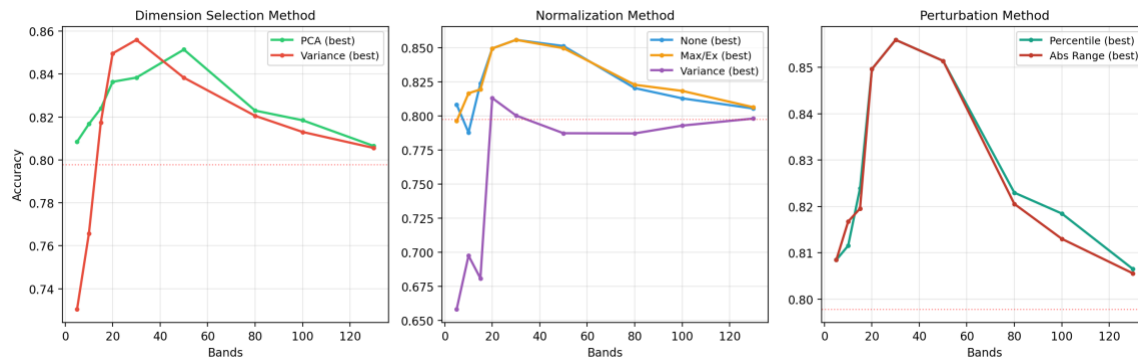


Figure 7. Parameter sensitivity analysis: best accuracy vs. number of bands for each dimension selection method (left), normalization (center), and perturbation method (right).

Key Findings

- Band selection exceeds KNN baseline at ALL tested band counts (5–130), with peak improvement of +5.8 pp at 30 bands (81% reduction).
- LDA with just 5 selected bands (97% reduction) achieves 84.6% accuracy, exceeding the KNN all-bands baseline by 8.3 pp.
- All 10 classifiers match or exceed their own baselines at 30–50 bands, demonstrating classifier-independent band selection quality.
- LDA with 50 bands retains 99.7% of its full-spectrum performance (92.5% vs 92.8%), representing the practical optimum for this dataset.

- Exposure time and lamp power normalization is critical — the raw data showed 100× intensity imbalance across excitations, which biased band selection without correction.
- The framework generalizes from lichens (distinct biological classes) to collagen (subtle chemical concentration differences), validating cross-domain applicability.